

Project Synopsis Template

1. Title

Sales Business Intelligence Dashboard

Working Project Title

Sales Business Intelligence Dashboard

Team Details

Student Name	College Name	Branch	Mobile No	Mail ID	Role (Lead/Member)
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Program Track and Domain

Program Track (Engg-HUB / Eng-Spoke / Science-HUB / Science-Spoke / Commerce-Spoke)	Domain (Analytics / Machine Learning / Deep Learning / GenAI)
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Commerce-spoke	Analytics and GenAI with Power BI
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Guide Details

Mention the name and designation of your project guide.

2. Problem Statement

Problem Statement

Organizations in the retail sector generate large volumes of sales data on a daily basis. However, this data is often stored in raw form and contains issues such as duplicate records, missing values, and inconsistent formatting. This creates a major challenge in analyzing the data effectively and extracting meaningful insights. This problem is significant because inaccurate or unclean data can lead to incorrect analysis, which directly affects business decisions, sales strategies, and overall profitability. Without proper data processing, organizations may fail to identify important trends, customer preferences, and high-performing products. The primary users affected by this problem include business analysts, managers, and decisionmakers who rely on accurate data to monitor performance and plan strategies. Retail companies, in particular, depend heavily on data-driven insights to stay competitive in the market. Existing approaches often involve manual data handling or basic tools that are time-consuming and prone to errors. Additionally, many systems lack interactive visualization and user-friendly interfaces for exploring data. There is also limited use of AI-based tools, such as chatbots, which can simplify data interaction and improve accessibility for nontechnical users.

3. Objectives

- To collect and understand the retail sales dataset containing 9994 records and 21 fields from Kaggle.
- To clean and preprocess the dataset using Microsoft Excel by removing duplicates, handling missing values, and correcting data formats.
- To perform data analysis using Excel and Power BI to identify key patterns such as sales trends, product performance, and regional analysis.
- To calculate important KPIs such as Total Sales, Total Profit, Quantity, and Profit Margin using Pivot Tables and DAX measures.
- To develop interactive dashboards in Power BI including Executive Overview, Sales Analytics, Risk and Performance, and Tooltip pages.
- To transform raw data into meaningful insights that support effective business decision-making.

4. Methodology / Implementation Plan

- **Data Collection:**
 - The Retail Sales Dataset was collected from Kaggle and contains 9994 records with 21 columns related to sales transactions.
 - The dataset includes customer, product, sales, and regional information used for business analysis.
 - Initial exploration was performed to understand the dataset structure, data types, and relationships between variables.
- **Data Cleaning & Preparation:**
 - Duplicate records and missing values were identified and removed to improve data accuracy.
 - Inconsistent data formats were corrected and proper numerical and categorical data types were assigned.
 - Columns such as Sales, Quantity, Profit, and Discount were formatted and organized into a structured dataset.
- **Data Analysis:**
 - Data analysis was carried out using Microsoft Excel and Microsoft Power BI to generate business insights.
 - Sorting, filtering, and pivot tables were used to analyze sales and performance trends.
 - KPIs such as Total Sales, Total Profit, Total Orders, and Profit Margin (%) were calculated for performance evaluation.
- **Dashboard Development:**
 - Interactive dashboards were created in Microsoft Power BI to visually represent business performance.
 - The dashboards included Executive Overview, Sales Analytics, Risk & Performance, and Tooltip pages.
 - Visualizations such as sales trends, product performance, regional analysis, and KPI summaries were included for decision-making support.
- **Insight Generation:**
 - Sales trend analysis was performed to identify high-performing periods and seasonal patterns.

- Product performance analysis helped identify top-selling and low-performing product categories.
- Regional analysis was conducted to compare sales and profit performance across different locations.
- Customer and profit insights were generated to support better business decision-making and strategy planning.

5. Resources and Timeline

A. Hardware & Software Requirements

Hardware Requirements:

- Laptop with minimum 8 GB RAM and Intel i3 processor

Software Requirements:

- Operating System: Windows 11
 - Tools & Technologies:
 - Microsoft Excel
 - Power BI
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B. Dataset

- The dataset for the project is available publicly.
 - Source: <https://share.google/rb8QXqT6vpcaMeIPB>
 - Dataset Description: The dataset contains 9994 records and 21 columns representing retail sales transactions. It includes detailed information such as order details, customer information, product specifications, sales amount, discount, and profit, which are used for business analysis.
 - Dataset Fields: Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, City, State, Postal Code, Region, Product ID, Category, Subcategory, Product Name, Sales, Quantity, Discount, Profit, Profit Margin (%).
 - Dataset Usage: The dataset is used for data cleaning and preprocessing in Microsoft Excel, followed by KPI calculation, Pivot Table analysis, and trend identification. The processed data is then used in Power BI for creating interactive dashboards
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C. Project Timeline

PHASE	TASK	TENTATIVE COMPLETION DATE
Phase 1	Data collection and dataset exploration	Week 1
Phase 2	Data cleaning and preprocessing using Microsoft Excel	Week 1
Phase 3	KPI calculation and Pivot Table analysis	Week 2
Phase 4	Dashboard creation and visualization using Power BI	Week 2

Phase 5	Documentation, report preparation, and project submission	Week 3
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6. Expected outcomes

The raw retail dataset is transformed into a clean, structured, and reliable dataset through preprocessing.

Data analysis helps identify key insights such as sales trends, top-performing products, and regional performance.

Power BI dashboards provide clear, interactive, and visually appealing representations of data, enabling users to easily understand KPIs like Sales, Profit, and Quantity through charts, graphs, and summary visuals.

The project highlights important business patterns, trends, and performance gaps, helping organizations identify strengths, weaknesses, and areas that require improvement for better strategic planning.

7. References

GitHub Link :

https://github.com/mattdelaune/PowerBI_Healthcare_Dashboard/blob/master/Hospital%20Inights%20Dashboard.pbix

Youtube Link :

https://www.youtube.com/results?search_query=power+bi+healthcare+dashboard+tutorial

Research Paper Link:

https://journal.ijresm.com/index.php/ijresm/article/view/2670?utm_source=chatgpt.com